APPENDIX

4

Interviews Supporting Legal Aspects of DNA Testing

As part of completing my research on legal aspects of DNA testing for Chapter 18, I was privileged to interview several scientists and lawyers. What they had to say was informative and insightful—and so their full responses to my interview questions are included here with their permission. Portions of these responses have been used to create content for Chapter 18.

INTERVIEW WITH DR. ROBIN COTTON ON BEING AN EXPERT WITNESS

Dr. Cotton is associate professor and director of the biomedical forensic sciences program at Boston University. She has over 20 years experience with forensic DNA analysis and served as the Laboratory Director for Cellmark Diagnostics from 1994 to 2006. She has served as an expert witness and testified in over 200 trials regarding DNA evidence including the O.J. Simpson trial in 1995.

How many times have you testified in court?

At least 220 times in 35 different states. I know that you are supposed to have an exact count, but I don't have it. I stopped counting when I left Orchid Cellmark in 2006 but have probably testified 20 times since then.

Have all of your expert witness testimonies been for the prosecution? If not, what is the breakdown of prosecution vs. defense?

Almost all of my testimonies have been for the prosecution. I have testified 2 or 3 times for the defense. Cellmark used to do defense consulting but for the most part with DNA results, you do not get called to testify for the defense—unless you are willing to go beyond what you really think or there is an important mistake in the testing.

The best thing you can do to help a defense attorney, if the lab has done a competent job in testing, is to tell the defense attorney what he should be hearing from the expert. I have seen transcripts where the testing was fine but what the witness said about the testing isn't so fine or more likely it is very imprecise and therefore it could be misleading.

Would your preparation and performance in court be any different if you were testifying for the defense rather than the prosecution? How?

No. You need to know what you are talking about in either case. Remember that you are there to explain the data. As an expert witness, you are not there to take a side. Some attorneys would like you to take sides. If it is a defense attorney, they may tell you all the things about the case that make it seem like the defendant is getting beaten up by the prosecutor. The prosecutor may tell you all of the bad things that the person did in this case.

There is no question that some of these defendants have big issues and have hurt people. Our role as expert witnesses isn't to participate in that discussion. If you are in a law enforcement lab, you are at serious risk especially if you have no process by which you can tell somebody that you are being leaned on by an attorney—because it can happen. It happened to me—and I was from a private lab.

What do you enjoy most about being an expert witness?

I am going to use the word "contest." I don't really like the win/lose ambiance that you experience in the courtroom. But sometimes the defense attorney or whoever is on the other side—but in my case it was mostly the defense attorney—really knows what they are doing. It is almost like you are having a really tough discussion. They are doing things that are a little bit hard to contend with—and if you can manage that well (so that you are managing to make your points well) that feels good.

It can be really boring. If you have said the same thing 200 times, sometimes it can be really boring—and you can't be bored! You cannot allow yourself to be bored. It is sometimes hard if it is a very routine case. If everything sounds so practiced, that you are not being spontaneous, then I would be concerned that the jurors would notice that. Remember that this is their only case. The jury will not see you again.

What is your role as an expert witness?

Your job is to represent the data. I think people forget that usually just one person signs the report. I think that it is important that many people in the lab would have the same opinion about that report. When you are sitting there on the witness stand, you are the lab! For purposes of that jury, you are the laboratory. Whatever they think about the laboratory comes from what they think about you.

Have you have ever taken one of those classes regarding risk management, where they talk about what you do not often do but if it fails, it is a catastrophe? Courtroom testimony is one of those things. Doing a poor job with courtroom testimony can cause serious problems for a laboratory and the analyst. If laboratory personnel are testifying all of the time, then every one of those people has to be good <u>every time</u>.

How do you make sure that testimony is consistent across your laboratory?

At Cellmark, we had two people signing every report so either one of them would go to court and testify to the results. The technical reviewer was also signing along with the analyst who did the initial interpretation. Thus, we had two people available to go to court because we were testifying all over the country. It also allowed us when there were *Frye* hearings to send the person with the more academic background. This approach gave us versatility. But we were not paired up. A case from any analyst could be picked up by any technical reviewer. This approach helped to have consistency across the lab. If we had something where they were not agreeing, we would bring in a number of people to look at it.

The thing that worries me the most is that a brand new person is being technically reviewed by another brand new person. They could reach an interpretation that others in the lab would not agree with and off goes the report. This would be a problem!

What did you least enjoy about being an expert witness?

Very aggressive, very loud, very unpleasant cross-examination tactics. I don't personally care to be shouted at.

How did you handle being shouted at during cross-examination?

Well, I had a big advantage when I started testifying. When I was a postdoc in Iowa, I did volunteer work for three years at a crisis intervention center. The experience taught me a lot about talking to people. You were talking to people that were very upset, very depressed, and very angry—and sometimes verging on out-of-control. We learned that if someone is shouting at you then keeping your voice down will bring their voice down. Keeping your speaking tempo slow will get their tempo to slow. I used those approaches in the courtroom to slow my pace. And I always remain attentive. If you are being attentive to the defense attorney who is yelling at you, it makes him or her look poorly if they are shouting at you. If I am sitting here being quiet, listening intently, answering slowly, and he is shouting at me, then he will not look good!

Have prosecutors ever tried to push you to say things that were beyond what you thought were appropriate?

Yes, there have been times that I was leaned on. I learned that a lot of prosecutors will say "can you make a stronger statement?" I think that is an okay question. But once I say "No!" then I do not think they should ask again. When giving court testimony, people need to know the limits of their data and stick to them.

How do you know what your limits are?

You need a thorough understanding of the literature and your protocols. If you cannot go directly to the source, you shouldn't be saying it. Unfortunately some analysts today don't read very much and therefore do not know the source information very well. So when they say it is "this" and you lean on them, they won't know!

The reason that it was so easy to do *Frye* hearings was coming from an academic background. You are used to understanding that there is literature to back this up, how do I find it, and what does it say.

How has being an expert witness evolved over the years?

After the Simpson case everyone decided it was a disaster to give a lot of explanation so now they give almost no explanation. I am not sure that that is not going a little far over the edge but it is hard to give a lot of explanation. For example—"capillary electrophoresis"—that is a little tricky!

Do you ever show electropherograms to the jury?

I don't, but I do show them the allele results table. The table isn't that hard to figure out. People should be showing the jury the table, it's not that hard. Here are the names of the loci and here are the samples and here is the data. If you have taken the time to explain something simple, like paternity, for one person each locus has two alleles—you get one from mom and one from dad. Then the jury sees the two alleles or more than two. You can say to them that the instrument data is more quantitative than the data in the table. So I can tell you if I go back to the instrument data, and determine whether there is more DNA represented in this "10" or this "11" and then you have made that connection. You can even draw it.

Do you use PowerPoint slides to emphasize a point or to teach a concept?

Never! I never do, but this is personal to me. I think that when you go to be an expert witness you have to think about how do you best explain things. If you love PowerPoint slides, then use them.

I had an important insight early on in my court testimony experience. When you are sitting in the witness chair, you only have your hands. An attorney in Seattle said to me: "Why don't you come down and diagram it?" It was like night and day. As soon as I stepped down and starting drawing, I was no longer confined to the rigid position of the witness chair. I was me. I was teaching! I was connecting with the audience. I could draw and if I didn't like it, I could undo it.

Is it a challenge that juries cannot ask questions and therefore you don't really know what they are thinking?

There are courtrooms in certain states where the juries are allowed to ask questions. They do it by writing the questions during the testimony and then when you are done, they pass the questions to the judge. The judge reads them and if he thinks they are okay, he will read them to you so that you can respond.

I don't think courts should be on TV. I think it is fine to have plenty of people in there but I think that having things on TV is not a good idea. I think that everywhere jurors should be allowed to ask questions. I know they can in grand jury. You can figure out immediately from the question asked how well they understood what you have said. Sometimes judges will ask questions, but that is more often in an admissibility hearing because they are the ones having to make the decisions. You can have two or three questions from a jury and realize that you didn't make something clear and you have another opportunity. It makes a huge difference!

Normally you only get one pass. I have been in the witness chair listening to what I am saying and thinking. "Are you out of your mind? The jury could not have possibly understood that statement. That's gibberish!" Sometimes it comes out well and sometimes not.

Do you ever read your transcripts to see what you said in a trial?

I prefer not to read my transcripts. I know that is bad on one account. Sometimes when you have to do a retrial they want you to read your transcript. But my feeling is that if I answered it the best I could the first time, I am going to answer it the same. If I read the transcript, then I am worried about whether I am remembering what I said previously. If I don't read the transcript, I have taken that concern off the plate. If the attorneys say, "Well, that is not what you said the first time," then I am allowed to ask to look at what I said the first time. Then I can say but that question was slightly different or I was answering that question about a different locus or whatever. As long as you tell the truth, you don't have to remember exactly what you said.

How do you prepare for trial?

I prefer to prepare the day before testimony. Now that does not mean that I might not look at the case a couple of weeks before, go over it and see if there are any issues: Are there typos? Is there anything I need to amend? Is there any problem that I need to be aware of? But the details, I don't do until the day before and then I outline the results. I just take the case folder apart as if I were doing that technical review all over again. I go through absolutely everything—from the evidence log-in sheet through every piece of paper so I know actually what's where. I may have the original case folder with me, but I have a copy of the whole case folder or the pieces of paper that are most important to me so that I can mark them up. Whenever you mark up paperwork and you carry something extra into court, you do have to be aware that you may have to turn that over. So I make my markings neat. I might highlight something if there are a lot of samples and you are trying to keep track of them, I will highlight the name of the sample. I want something that will draw my eyes to exactly where I need to look. I will put an arrow over certain peaks or I will use stickies. Whatever I do, I do it neatly. I work such that if I have to give it over…I am not going to be embarrassed by what I have written or marked up. When I study this way, the information that I need in preparing for trial stays with me better. I also know that I will not be fumbling around with the case folder.

We used to compile our case folder with the first thing being the bottom page and new information just got piled on top. For court, I would disassemble the case folder and I would put all of the pages for one sample together if there were many samples. If there were not many samples, I would organize the pages by the step: here are the extractions, here is the next thing, and so forth. So I would reassemble them in an order that made sense to me—at least in my copy so that I could work from it easily in the witness stand.

I talked to Gary Sims [from the California Department of Justice DNA Laboratory] after the Simpson trial. He did his whole testimony from memory. I can't do that. It is a big risk if you try it and are not good at it. The risk is that you will say something wrong. I use the paperwork when I need to. I look down at my notes while I am talking if needed. You sometimes have to ask permission to do that or the attorney will say, "Do you need to look at your casework to refresh your memory?" and then you are good to go. I would much rather see the jury see me rely on the paperwork than say something and go "Oh, my! I misspoke. It wasn't that piece of evidence, it was this piece."

There are a couple of things that I think the juries aren't going to worry about. Number one, who in their daily routine doesn't write lists or have to look at pieces of paper. The

other thing is that expert witnesses get worried a lot when there is a mistake. A lot of times the mistakes are little, but occasionally the mistakes are big. But if you have to say you made a mistake, I don't think that it wrecks your credibility or your testimony because everyone sitting in that jury—whether they have a high school education or a Ph.D.—makes mistakes. And to think that you would not make a mistake is unrealistic. It's embarrassing, it's hard. I had to testify to early mistakes in proficiency tests at Cellmark for 15 years—and every time it was hard. It doesn't get easier to say it, but you have to say it. There is nothing else to do so you might as well get comfortable with the idea that it happened and get comfortable with the idea that they are not going to hold it against you.

What was the most challenging trial you were involved in and why was it so challenging?

There were several. One was the O.J. Simpson trial. There was another one in Colorado where the defense attorney was just the most aggressive, most angry, most mean, that I have ever experienced. He raised his voice all the time, he insinuated that I was a liar, that I was unethical, and untruthful. And it is very hard. It is hard not to get angry, it is hard to keep your cool, and it is hard not to give in when you are being badgered. The Colorado case was not the length of the Simpson trial but still difficult especially since the prosecutor wasn't terribly helpful. He didn't object and the judge just let it happen. It is difficult when a judge is not trying to exercise any control.

How do you prepare yourself for testifying in court?

You have to be really focused in the courtroom. You cannot mentally be somewhere else. You need to listen to the questions. The longer you are listening to questions, the harder it is to stay focused. Just like anything, you get tired. The attorneys get to take a little rest, they trade off, etc., but you cannot get tired. You have to be very focused. You cannot be paying attention to everything in the courtroom. It will distract you. I just look to the attorney and I just look to the jury and that's it.

How do you dress when you go to court?

I wear dark clothes. I wouldn't choose a red suit. I don't want to look like I am going to a party. Nobody in that courtroom is there because something good happened. You have to respect that things have happened and you have to have respect for the proceeding and for the individuals that are there.

Do you have a routine you follow as you enter a courtroom?

The minute you walk into the courthouse you need to be aware that anyone around you could be a juror, an attorney, or someone else involved in the trial. Your professional behavior starts when you walk in the door and you don't change that until you are far away from that courtroom. You are not pals with the police officer. You are not pals with the attorney. You are not there to be somebody's friend.

Are there any specific things that you bring with you or do when you sit in the witness box?

I take whatever I think I am going to need to the witness box. If there were a *Frye* hearing, I might bring in scientific articles. For a case, I might bring some notes along with my copy

of the case folder. I don't have things with me that I'm not willing to turn over. Some people take a book if they think they might be waiting—and that is fine. I usually take some scientific journal articles just because *People* magazine doesn't seem to be the right thing to have in a courtroom.

How much time do you normally spend talking to the lawyer who has called you to testify before your appearance in court?

It really depends on how much the lawyer knows. Sometimes you have to force the lawyer to talk to you, and that can be very hard for analysts from crime labs. At Cellmark, as a private laboratory, we had a way to get the lawyer's attention. We would say, "I am not getting on the plane until I know we are going to have time to sit down and talk!" An analyst in a public crime lab may not have that option but if there are issues, your lab director should meet with the district attorney and work on improving relations between the lawyers and the lab.

Do you have a list of prepared questions that you share with your lawyer before and encourage him/her to use while you are on the stand?

Lawyers like it when you have a list of prepared questions for your direct examination. However, you need to go through the questions with them because some of the questions may not fit a particular case or your role in the expert testimony you are offering.

What is the most important thing for the jury to know about you when you are explaining your qualifications?

I don't know. It's the worst part of the testimony. I have never been comfortable with it. I actually feel embarrassed when I have to say my qualifications because I know the people in the jury may not have had as many educational opportunities. So I just say my qualifications and hope to move on quickly.

What should be included in a CV (*curriculum vitae*) that is submitted to court as part of being qualified to be an expert?

Your CV should be only the organizations you are a member of, your publications, your job title, your job description, and your education. Testimony is not a qualification, neither is your volunteer work or anything else. I am careful about what I am qualified in. For example, I never allow the attorneys to qualify me as an expert in statistics, but they may qualify me as an expert in the calculations of frequencies as it relates to the DNA testing because I know how the frequencies are calculated.

How do you know whether or not you are connecting with the jury?

You can just forget it. You don't know. Of course it's important to connect with the jury. I want them to understand. I'm choosing words carefully. When we changed from doing RFLP testing to PCR testing, the first time I went to court I realized that I hadn't figured out what to say yet. So I sat down and wrote out the absolute basic pieces of information that a person has to understand in order to understand what I am saying. Those are the things you want to talk about in plain language.

Do you use analogies to help explain DNA results to the jury?

I am not an analogy person, but some people are. An analogy that I almost always use is that the variation in DNA repeats, which gives size differences in the DNA between people, is similar to the number of boxcars on a train. More boxcars lead to a longer length train. It can be hard for people to envision DNA so another analogy that I have used in the past is to say, "Suppose I line up on the table here in front of me 46 spools of thread, two blue, two orange, two green, and so on. Then we unroll the thread from the spool and that's the DNA."

Some people are very fluid with analogies. Their analogies are simple and they use them well. Other people think they are very fluid with analogies, and the analogy is so complicated that nobody can figure it out! You have to be careful about using analogies because you don't want to talk down to the jury too much or too often. If you use an analogy for everything, the jury may get tired of it. In addition, they won't know which analogy went with what, which is even more dangerous.

What do you want jury members to come away with after your testimony?

If you are going to talk about DNA, you have to know something about the fact that people have chromosomes, and each one of those chromosomes is a piece of DNA. For each chromosome pair, you get one from your mother and one from your father. You can work out what is the most simple thing they have to know and then how to say that using the simplest possible language. You quickly find out that there are a few scientific terms you can't do without. You can't do without "allele." You can call it a "type" or you can call it whatever, but you have to have a word for it that you have defined. Make sure you have as-plain-as-day explanations for the scientific words you use.

It used to be a lot more fun when you could make a good joke about Southern blots—because you are sitting there talking about DNA and it is supposed to be really high-tech. Then you talk about having this gel that is sort of like Jello and you are putting a big stack of paper-towels on it like a big weight on top and the jury is thinking, "How high-tech is this?" There are places you could have fun. The CE (capillary electrophoresis) doesn't quite do as well.

Were there ever times when you felt like the jury did not understand what you were trying to explain?

Oh, absolutely! After listening to your explanation of a topic and seeing the blank look on the jurors' faces, you may need to say, "I don't think I got that out right. Can I try it again?"

What do you find to be the most difficult thing to explain to a jury?

It is hard to explain a differential extraction to a jury. I think it is easier to draw something as a diagram than to sit in the chair and explain it with your hands waving in the air. A PowerPoint slide can do the same thing for you. However, the jury does not see you create the slide. If I stand up in front of them and diagram the concept I am trying to teach, they are seeing that I am making this out of my head. I think you connect better with them, plus they know that you know what you are saying if you are drawing the concept in front of them. I can put a lot of information on a PowerPoint slide and not really own it, but if I am drawing it for them I have to own it.

Sometimes if I have been given a difficult question, I will say, "I can explain this to you, but I can't do it without a diagram." As part of preparing for my testimony, I try to remember to tell the attorney to bring an easel and paper to the courtroom so that I can be prepared to diagram an explanation.

Are there any props or handouts that you regularly use to help explain DNA to the jury?

Some attorneys like props and others don't. I had one really bad experience where I went to a trial, and the attorneys had made a lot of posters—and had not just taken the table from the report. They had made posters with pictures of the evidence, which was kind of gruesome. I was very uncomfortable with that.

Is there an ideal direct examination in your opinion?

No, because that's just so case-specific. It could make sense though to have questions prepared to help you logically flow through the information being presented.

What are your thoughts about how to handle discovery requests? Do you push back if you feel like too much information is being requested by the opposing counsel and experts?

I think the most important question for a lab (in fact, any discipline in the lab), is if you were to hand this information—the information you are going to give over to the defense—to a respected colleague, would you have given them everything they need to evaluate your work? If the answer is yes, then that's about what you should be giving on discovery. For DNA that might be your case folder, your procedures, and your electronic data. I know that the defense often asks for lots of other documents: validation studies, proficiency test results, etc. Different states have different rules and different prosecutor's offices have different attitudes about giving over that extra material.

I don't see anything wrong with asking people to pay for getting a copy of the discovery information. At Cellmark, we used to get huge requests. We found out that the nicer, neater, tidier, and more organized the paperwork was that we gave to them, the less we ever heard about it again. It was almost as if they looked at it and said, "Oh my! This information is organized. It must be good."

In any case, you should give over, *without question and happily*, the things that you would need from another competent DNA person to perform the review. *Do not worry about who might look at the information as a defense expert!*

The defense attorney can take it to any defense expert. The defense attorney isn't necessarily going to control what this expert does with your data, but hopefully the defense attorney would try to recognize whether or not their expert was making a legitimate argument with your data. If the reviewing expert does not make a legitimate argument with your data, then that is their problem—not yours. Unfortunately, I find a discovery request can become the start of an "us" against "them" mentality.

You know what you did and you know what your standard operating procedures are. You know hopefully why you did not choose to look at peaks below a particular RFU [relative fluorescence unit] level. If their review of your data makes you have to study why you are doing what you are doing, then that's part of your job, too. You wrote a scientific report. Your data should be outlined in the report. The table should be in the report.

The stats should be in the report. That is the basis of your scientific opinion. However, it is the case folder that holds the entirety of what you did; and it is the SOP that says how you did it. So let the defense attorney have all of this information! If I were reviewing results from a case, I would ask for all of this information.

Have you ever done an affidavit instead of testifying in court?

I have done affidavits, but I have never done them as a replacement for testifying. I know that sometimes the prosecutor and the defense have stipulated to our report and just put it into the court record. Maybe the DNA testing wasn't particularly probative, but they just wanted to show that they did it. However, this is different than writing an affidavit.

Are depositions more challenging relative to testifying in court?

Depositions are more challenging because there is no judge, so the defense attorney can ask you something really obnoxious, such as, "How much do you make?" They can ask personal questions and there is nobody there as an arbitrator. While that doesn't happen very often, I have had it happen—and I've been forced to answer. The prosecutor will say "I object" and that goes on the record.

What are the lawyers like whom you have met in your court experience?

You learn that some lawyers are fabulous and some lawyers are awful—like a regular bell curve. You can learn so much from a good lawyer. A really good lawyer will (a) not press your testimony, and (b) can give you coaching that you are not going to get anywhere else. I don't mean coaching in what to say. I mean coaching in how to deal with a difficult or badgering defense attorney.

What was the best coaching you ever received from an attorney?

The best coaching I ever received was from an attorney in Montana years ago. He was one of two attorneys on the case in a *Frye* hearing and I was one of several witnesses. We were watching other experts and he was commenting on them. He taught me that you can say too much when trying to answer a question. I had not previously realized this because in my efforts to be helpful, I was giving people a lot of information. The defense attorney in that case was very good with his questions and every time a witness would go beyond the question with additional information, the defense attorney took the information the witness had told him and dreamed up another question on the spot. It was miraculous. The person I was with was from the Attorney General's office, and had many years of trial experience. For example, he taught me that if you would just say in response to a question from the defense attorney, "No, capillary electrophoresis (or whatever the question relates to) does work very well. There is scientific literature to back that up!", the defense attorney does not know what the scientific literature is and so you are not giving him something to ask about next—except maybe "Have you read the scientific literature?"

When I got up on the witness stand, I tried this approach—and it was helpful! As soon as I started providing short responses without the elaboration which he had not requested, the defense attorney had less to say. He was remarkable at thinking of the next thing to ask from a witness's previous answer. But when I just said, "No, that is not correct" without any further explanation, he didn't have anything to build on. The best thing I ever learned was

to say less and just answer the question—and only the question! I am not withholding information. I am just not giving explanations that were not requested.

How do you handle situations where the opposing attorney is attacking your testimony?

When somebody is attacking your testimony, there are two bad things that happen. One is that you get mad. The other thing is you get tired so you tend to start giving in or you just shut down or you glare at everybody and you look uncomfortable. It can take some practice at home, such as rehearsing how you might respond to challenging questions.

If the opposing attorney is just attacking you, and it is a really small point and they are correct, just say "You're right!" Sometimes they're making a point and they are correct. However, our natural tendency is to think "I'm not giving in or giving up on anything." You have to let go of that.

If the opposing attorney finds a typo in my report or case folder, I just acknowledge the point. I do it in as easy a manner as possible. Sometimes they are trying to make a point that's kind of partially right and that's the hardest thing, because they want to make it simple but you know scientifically that where they are going is not correct.

Do you try to think about where they are going with a particular line of questioning while you are answering questions?

You have got to be careful with trying to think about where they are going to go. Because (a) you can get very distracted by that and, (b) you don't really know. It is very easy to think "Oh! I see where this is going." Most of the time you may be able to see where a particular line of questioning is going but sometimes you will not. My advice to people is to just focus on the question at hand. If your mind is five questions away, you will not be paying attention to the question that is in front of you.

Sometimes questions get very hard and you can't always just manage the situation by saying that it is more complicated than the way they have described something. You may be faced with agreeing with something that you don't totally agree with or saying it is wrong when it is not totally wrong. That to me is the hardest thing to deal with because you have to weave your way through a series of questions while trying not to be an obstructionist or giving in. Sometimes you just have to say "It's more complicated than that but under this assumption it would be true…"

Do you ever restate a question to try and help the attorney?

It depends—most of the time you are just correcting their question. For example, they may have just said to you, "The RFLP technology used in this DNA typing procedure is new or novel, isn't it?" So you go, "No, this isn't RFLP, this is PCR?" Their verbiage is all wrong, sometimes just out of kindness or not to be totally obnoxious you are fixing the question and answering it.

Be respectful of the attorneys. They know a lot of law that you don't know. It is just that they don't know the DNA information as well as you. Sometimes they have pre-prepared questions that came from years ago so the vocabulary is incorrect. Attorneys for some reason cannot capture the word "locus." You don't want to correct them too much—but you can help them a little bit. If it is a cross examination, and they are being not very friendly, then I'm careful about what I add.

If the attorneys on both sides are allowing you to answer the question and you are not taking advantage of that and talking for 10 minutes, then you can usually say "Well under this circumstance that's true, but sometimes under other circumstances it is not…" But if there are a lot of objections, and there has been antagonism between the two attorneys during the trial and you know that the defense attorney is going to say "Non-responsive!," then you can feel pretty stuck and you just have to figure it out as you go.

Have you ever felt that the side you are testifying for is not supporting you—that you have been abandoned?

It's not so much that. It's all about the lawyer's skill. I have walked into a courtroom with a lawyer who was so good—and we did not do very much pretrial work—but he was so in charge and so careful to listen. I felt completely safe even when it was somewhat adversarial. I have also felt completely like neither the judge nor the attorneys were going to be helpful. It may be they don't know what to do. It may be that they are out of their league. It could be any number of things. Then you just do the best you can.

What is the worst thing that you can do on the witness stand?

I actually think that there is only one mistake you can really make—other than telling a lie, of course. You should not treat the prosecution and defense attorneys any different in how you respond to them. However you talk to one attorney, you need to talk to the other attorney the exact same way. You don't want to change your body language. You don't want to change your tone or the speed in which you answer questions or your demeanor to the jury.

What is your attitude toward defendants in a trial?

Respect for everyone in the courtroom is important. I always refer to the defendant as Mr. Smith—never as "the subject" or "the suspect." As far as I am concerned until that verdict comes down that person is innocent.

Early on in my career a prosecutor taught me that in her view individuals were being prosecuted for an event, which did not necessarily define them as a person. Defendants are human beings. This important lesson from this prosecutor gave me a much better perspective.

What is your attitude toward defense attorneys having faced more than 200 of them?

Some defense attorneys completely understand their role and they are able, like a good prosecutor, to pull it off without avarice. They can be hard. They can be tough, which is not the same as being mean and angry. You can really tell the difference between the two. Defense attorneys have their role. Often, the first thing that new people think, which I thought too, is "Oh my! How can you defend this person who serially raped 25 women?" Well, those defendants have a right to their day in court and deserve the protections that the law provides.

What is the best advice you have received from an attorney?

I had a couple of very good attorneys to work with early in my career of testifying. They spent the time to teach me because the DNA data was so important to their case. I received

lots of on-the-job training from some very skilled people—both defense attorneys and prosecutors—because the early DNA cases were all admissibility issues, and people weren't spending the money unless it was an important case. With an important case, you generally have very good lawyers on both sides so I was privileged to have worked with these very good lawyers early on, a bunch of them.

In one of my early *Frye* hearings, I was stressed about whether or not the judge would find the DNA results admissible. The defense attorney was difficult, and I could feel that things were not going well. Sensing my anxiety and feeling my sense of responsibility for the outcome, the prosecutor looked at me and said, "I win or lose this case – not you!"

How do you act around the lawyers and other participants in the courtroom?

I may be friendly with the lawyers in their office but not inside the courthouse. I would never be unfriendly to a victim or a victim's family, but I also don't want to have a conversation with them in the courthouse. If they say something to me, I will answer the question quietly, but I won't engage in a big conversation with them. I'll just be polite—just like I would if the defense attorney asked me a question. *Equal respect to everyone…*

At the crisis intervention center where I worked while a postdoc, they had a phrase for this behavior: "unconditional positive regard." That should be your attitude to anyone in the courtroom. If I have a private thought about some player, then I can articulate it when I come home. I can articulate it with the attorney I am working with in his office. I can go back to the lab and talk to a colleague, but in that arena of the courtroom, they all get my equal respect genuinely. I think that is where the creditability comes from that you should have as an expert witness.

What do you say when you are not certain of an appropriate response or not certain you have the right answer?

Depending on how uncertain I am, either I say "I cannot answer the question because I'm not sure" or if I am reasonably sure but not positive, then I will say "I think the answer is such in such, but I'm not sure. However, I could find out where to get the answer."

Have you ever made a mistake in your testimony and then realized it afterwards? How do you handle this situation?

I have made mistakes in testimony. I usually realize it during the testimony and simply say, "I realize I just misspoke." If you make some real mistake, then you have got to tell somebody. Don't try to cover it up.

What do you say to defense experts who seem to exaggerate claims or concerns about the data?

I don't think that a defense expert should make statements that are not scientifically appropriate. I don't think that a defense expert should insinuate that people aren't ethical when they have no history of that. I don't ever try to use that kind of heavy language. This type of behavior brings drama into the courtroom when there really should not be drama. The prosecutor or defense attorney may want the trial to turn into drama—but the science is better without it.

How do you deal with what can sometimes be harsh cross-examination tactics?

No one can really protect you except you. The minute angry or sarcastic words come out of your mouth then you have lost control. It is hard to get it back. It is almost like you have to hear the question minus the tone, minus the face, and minus the waving of hands. You have to train yourself to hear it like that. You have to not let the opposing counsel's behavior bother you. You have to be able to let go of the anger or not have it.

It is very hard not to take a tough cross-examination personally. Anytime you feel like you are being attacked then your anxiety goes up, your adrenalin goes up, and you feel like you have to defend yourself. I think that is just part of the tactic. You are not being attacked really. You are there just to say, "This is the data." And the defense attorneys are there to say, "Are you sure, did you do it right?" That is their job. If you didn't do your work right, then they need to know! It is their job to discuss anything that may be incorrect in what you did. They want and need to point it out.

How do you feel after having your testimony challenged in court?

Lots of people don't like to talk about how they feel, but if you don't talk about how you feel, then it is hard to change. You may talk about what the defense expert said or didn't say—and how they said it or didn't say it. You may begin to think that how other people treated you is the problem rather than "I'm my problem!" My feelings are my problem. Defenses attorneys can say anything in challenging your testimony. You cannot change this or fix this; it is part of the trial experience.

I learned how to fix my feelings of frustration after participating in the courtroom drama by remembering that it was a show. *As an expert witness, I have a very important role in this show—and the show has very important consequences—but the show isn't my permanent world*. I have strengths and other things outside of this arena. Whatever happens in the courtroom, I don't have to carry it out with me.

Having this perspective doesn't always keep you from feeling upset—but I don't have to carry those feelings of anger or frustration out of the courtroom with me, and I don't have to lay that anger on somebody else. I can just say, "You know, that was hard!" I can say, "I think that attorney, that argument was ridiculous!" But I don't have to get all riled up.

How do you handle questions outside of your expertise?

You just tell them it is outside your expertise and you can't answer them.

What advice do you have for DNA analysts who have never testified in court before?

Go explain it to your mother or father or brother until they understand. It does take a lot of practice to say it without using your normal scientific terms. Sometimes if you get tired on the witness stand, then you find that you are drifting back into the scientific language because it is easier.

Have you ever taken a court testimony workshop? If so, what were some valuable things that you learned from such training?

Yes, I have taken testimony workshops, but not very many. It is like other workshops that I have taken—you get small bits of information. Even now, when I sit in on a moot court class I learn things or I get reminded of things. So you can always get better.

Could the U.S. court system be better? Would better training lead to better treatment of expert witnesses?

You can't change the defense attorney's role. That's not going to work. Prosecutors should not lean on people to provide testimony that they would like to hear. That would help.

There is a lot of talk about ethics today. What are you opinions on the importance of ethics in courtroom testimony?

Well, it is obviously important because you are under oath. You are giving people information that is important and may have an impact on other people's lives.

What advice do you have for DNA experts in terms of being impartial as a witness?

You state what the data says or what it means. Then you are impartial because you haven't spoken outside of what your scientific testing has told you. If you think you know what your role is, then you had better make sure that is what your role really is. You are really there to explain and speak to what that data says. If you keep thinking about that, then you realize you are not there to assist the prosecutor or the defense attorney. The data is what it is—maybe messy, maybe not. This is science.

What is the single most important thing to remember when serving as an expert witness?

You are the voice of the data! You are not a voice for the victim, which is what some prosecutor's describe their role as. People will talk about the "prosecution team." I really don't think you are on somebody's team, but that isn't to say you wouldn't assist them in understanding or assist them in answering questions so that the data is clear to the jury. Just because you help in appropriate ways, it doesn't mean you are on "their side." If you want to feel like you are on someone's side, then you are welcome to feel so. You just cannot behave like you are on their side. You cannot let that feeling influence how you behave, how you speak, and most importantly, how you look at the data. I think it is a hard issue because you want to please the people you are working with, right? It is the prosecutor who is supposed to worry about the consequences of the trial. If you represent the data accurately in a scientific sense, then it is hard to go wrong.

INTERVIEW WITH DR. CHARLOTTE WORD ON ADMISSIBILITY HEARINGS

Dr. Word is currently a private consultant after working for more than 20 years with Cellmark Diagnostics. She has testified in numerous admissibility hearings and in over 200 trials regarding DNA evidence. She also serves on the editorial board for the Journal of Forensic Sciences.

How many times have you testified in court?

I have testified about 300 times in about 25 states from 1990 to 2010. A large number of the testimonies were in admissibility hearings (RFLP, DQα/DQA1/PM, CTTA silver stain, D1S80) with most of the remainder in trials. I do not believe that I testified in an admissibility hearing for fluorescent STRs using capillary electrophoresis as our lab

[Cellmark Diagnostics] only had a few cases, maybe two, requiring admissibility hearings since we performed CTT for many years prior to using the newer technology.

Is there a difference in preparation for an admissibility hearing vs. preparation for trial testimony?

Yes and no.

Yes, because just as in preparation for a trial testimony, the expert witness and attorney in an admissibility hearing must work closely together to prepare and present a series of coordinated and organized questions and answers to educate the judge about the issues and the case. And yes, just as in a trial, the expert witness is still a neutral witness in an admissibility hearing. He or she is an advocate for good science and good procedures only, not an advocate for the prosecution or defense.

No, because unlike in trials, the expert witness will often need to take a very active and involved role in the preparation for an admissibility hearing. First, it is important that the expert witness and the attorney have a good grasp on the legal issues of general acceptability controlling the hearing so that the appropriate supporting documents can be provided and foundation laid to assist the judge in the final ruling. Most states in the US are bound by either the *Frye* or *Daubert* cases, and some states have additional cases which further define the requirements for admissibility in that particular state, for example, *Kelly* in California and *Davis* in Maryland. Since admissibility hearings are rare these days, the attorneys often need to research the relevant cases, and the witness and attorney may need to network to get guidance from other attorneys and/or experts more experienced in these types of hearings.

Second, unlike trials where only the final report and/or evidence may be admitted during the proceedings and the attorney generally decides what should be admitted in a particular case, in admissibility hearings the expert witness will often have a big responsibility to assist the attorney in deciding what documents should be provided to the judge in the hearing as well as procuring and copying the necessary documents. Some of the documents that may be needed in addition to the case file include: primary research scientific literature underlying the techniques being used as well as literature showing the application to the forensic sciences, developmental and internal validation studies, reference lists, training manuals, publications and presentations from the laboratory, laboratory SOPs, glossary, list of court cases where the test results were previously admitted, and rulings from other cases. The expert witness may also have to help the attorney find other experts to testify regarding the general acceptance of a technique.

Last, unlike many trials, there is always at least one opposing witness testifying in admissibility hearings. The expert witness will often need to help educate the attorney seeking admissibility regarding the opposing witness' issues and opinions so that the attorney can prepare cross-examination questions to ask the opposing witness.

Is there a difference in the actual testimony in a hearing vs. in a trial?

Yes, definitely. Science, science, science, is what it is all about in an admissibility hearing. Sometimes the specifics of the case are not even discussed, or very minimally, in these proceedings. My briefcase is often packed full of papers and other documents that I may need

to refer to in a hearing—far more than what I routinely take to a trial. I especially like the well prepared, logical, and practical manner in which the questions are laid out to best educate the judge in the scientific issues. Since no jury is present, the testimony often proceeds with fewer interruptions and less drama.

Testimony is given while looking at the judge, which can be quite difficult sometimes if the judge is seated above and behind you. Finding a comfortable position to look at the attorney asking the questions, see the judge and have the court reporter hear you can be challenging. And sometimes the judge is so busy taking notes that you end up testifying to the back of his or her head and never make eye contact! Admissibility hearings also offer a great chance to see and hear other experts testify. This is a terrific opportunity to learn different approaches to presenting the science and how to be a better witness. You can also learn first-hand about the challenges of opposing experts.

Do you prefer testifying in a hearing vs. testifying in a trial?

I like both for different reasons. In an admissibility hearing, the testimony is all about the science. I get a chance to really describe the literature and techniques in detail, maybe even draw pictures. Since the judge often asks questions to clarify anything, it is possible to have a good handle on what the judge "is getting" and what areas still need more explanation and testimony. It is very rewarding to know, based on the questions asked, that the judge has a good grasp on the science and the relevant issues. I also enjoy working closely with the attorney during preparation and testimony.

During trials, the testimony is mostly about the laboratory work done in a case, the test results, and the conclusions. It is always fun to present the nice work from your laboratory. Often the jury is quite engaged, and it is possible to tell that they understand the testimony when they nod, smile, or grimace while silently answering the questions on cross-examination before I get a chance to answer them aloud!

What advice do you have to offer experts regarding testifying?

Be prepared! The more prepared you are, the more information is readily available to you and thus you will be more confident. A well-prepared witness who has carefully thought about the science, the case, opposing challenges, and possible issues will generally be able to handle any question asked by either attorney or the judge and can comfortably avoid answering a question outside of his/her expertise.

How do the questions asked in hearings vs. trials differ?

The questions qualifying the witnesses are largely the same from hearings to trials. And the general preliminary questions of: "What is DNA?" "Does DNA differ amongst individuals?" "What samples can be used for DNA testing?" etc., are also basically the same. Then the differences start. In the admissibility hearing, since the legal burden is to demonstrate to the judge that the scientific technique being used is a generally accepted and reliable technique in the scientific community and also that the technique works appropriately when applied to the forensic science arena, the line of questioning introduces the basic techniques, the necessary and critical steps of the procedure, and how this technique is routinely used in the general scientific community, for example, in clinical laboratories, research studies,

biomedical applications, environmental or industrial applications, etc. Particular examples may be introduced; for instance, in admissibility hearings for the introduction of DNA testing, it was common to discuss the use of DNA testing in human cancer diagnostics, organ transplantation, paternity testing, identification of war dead, identification of microorganisms in the environment, and so on. Primary research publications are often provided to the judge to demonstrate the use of the technique in a few of these "basic science" and non-forensic applications.

The next line of questioning introduces the foundational studies for the application of the technique to evidence analysis and verification that the technique works on samples traditionally analyzed by forensic scientists in crime labs. During these questions and testimony, the initial presentations at meetings and publications of the techniques and developmental validation studies are discussed and copies are generally given to the judge. Additional information regarding the validation studies performed in the testing laboratory and the development of the laboratory SOPs is often presented, and sometimes a copy of the lab's SOPs is provided to the judge. Publications and/or presentations from other laboratories using the technique may also be introduced. The reliability of the testing and any limitations of the technique would also be discussed during this part of the testimony. This is a good place to introduce safeguards in the testing procedures, quality control measures used by the laboratory, and to discuss other areas such as training, corroboration of results with other laboratories, proficiency testing, etc. The attorney may also ask you to identify other court cases where the technique has been admitted and discuss any cases where the technique was not admitted and your understanding of the reasons the judge did not admit the technique. Having knowledge regarding what other laboratories in the US and around the world are doing and in what jurisdictions the testing has been admitted can be very helpful here. Some judges will ask questions to clarify points and enhance their understanding. As I said earlier, this is an excellent way to gauge whether your testimony has been clear and educational or completely confusing!

Depending on the scope of the hearing, the discussion regarding the actual testing and results in a particular case may follow. In some hearings, this phase is omitted, while in others it is the primary focus of the hearing. This testimony is generally very similar to what is routinely presented to a jury during a trial; however, the answers may be more in depth to demonstrate the uses of the technique and the reliability of the testing.

What kinds of challenges are seen in admissibility hearings?

In most situations, the challenges are the same seen routinely during trials, but with more emphasis on the unreliability of the technique or the limitations of the test system for forensic samples in general. The "newness" of the technique to the analyst and/or to the particular laboratory is often discussed with the focus being that something new must not be trustworthy and that the lack of extensive prior experience must lead to a higher risk of mistakes. As in trials, often hypothetical situations and "what if" questions are brought up. The specific challenges introduced will largely reflect the scientific background and knowledge of the expert testifying for opposing counsel. Reading any reports, affidavits, declarations, and/or prior testimony from the expert testifying for opposing counsel can be very helpful for understanding in advance what types of challenges to prepare for.

We do not hear much about admissibility hearings for DNA testing these days. Why is it important for DNA analysts and attorneys to know about how to do an admissibility hearing now?

That is true. Although there were huge challenges to RFLP (restriction fragment length polymorphism) testing in the late 1980s and early-to-mid 1990s and to PCR testing using DQA1/PM and silver-stained STRs in the 1990s, by the time fluorescent STR testing with capillary electrophoresis became the predominant DNA testing technique worldwide, many of the basic issues of forensic DNA testing and interpretation had already been addressed by the courts at the trial and appellate levels. In the past decade, there have been very few significant challenges to the introduction of fluorescent STR DNA or mtDNA testing in the courts, so most analysts and attorneys have no or limited experience with these types of hearings.

However, I think it is still important to understand what an admissibility hearing is all about so that one can be prepared if challenges to the testing procedures are raised. What I see most often happening today is what we can call "pre-admissibility hearings" that are being held to determine whether there is a need for a true admissibility hearing. The underlying question at these hearings is whether there has been a significant enough modification to the basic and current generally accepted forensic DNA techniques to make it necessary to have an admissibility hearing on the modification(s). I think this brings up a very interesting scientific and legal question—when is a modification to the generally accepted procedures different enough to warrant a new admissibility hearing? Adding a new STR kit, capillary electrophoresis instrumentation, or software has not usually required a hearing. A few recent proceedings that I am aware of have focused on the testing, analysis, and interpretation of samples having, or allegedly having, small amounts of DNA (low-template DNA, or LT-DNA) that have been tested with or without modifications to increase sensitivity. And as is always possible for a variety of reasons, there can be hearings to rule on the admissibility of the test results of certain items of evidence in a specific case.

INTERVIEW WITH TED HUNT, J.D., ON A PROSECUTOR'S PERSPECTIVE

Mr. Hunt is the Chief Trial Attorney at the Jackson County Prosecutor's Office in Kansas City, Missouri. He has been a prosecuting attorney for 20 years and tried over 100 felony cases, most of which involved the presentation of DNA evidence. He is also actively involved in training attorneys and law enforcement officials in forensic DNA-related litigation for a number of state and federal organizations.

What is your role in the criminal justice process?

The prosecutor's role in the criminal justice system is to seek justice in each case rather than to merely convict. In many instances, seeking justice equates with the aggressive prosecution of a criminal defendant. In other cases, however, seeking justice might mean dismissing a case because the evidence of guilt is in doubt. It may also mean reducing a charge when the evidence reveals that a defendant's degree of culpability is less than previously believed.

As a lawyer, how much do you want or need to know about the details of DNA testing?

As a prosecutor, I want to have a solid working knowledge of all aspects of forensic DNA testing. However, that doesn't mean that I want to orchestrate a courtroom class on forensic genetics or analytical techniques. I believe that the true value of being literate in the field of DNA analysis is to provide myself with the opportunity to make an informed choice about what topics and questions should be included, and perhaps more importantly, *excluded* from my courtroom examinations of both prosecution and defense experts.

Aside from a strong working knowledge of the biological, technological, genetic, and statistical aspects of forensic DNA analysis, I believe it is also important that prosecutors have a firm understanding of the quality standards and guidelines in the field. It's very difficult to follow the "game" without knowing the "rules." These criteria provide a benchmark against which the overall work product in each case is judged. Confidence in DNA test results can be enhanced by prosecutors who familiarize jurors with the rigorous quality processes and procedures to which the analyst's work was subjected. Alternatively, jurors can rightly be made skeptical of a system or an expert who failed to follow community-wide quality standards and guidelines.

What resources have been helpful to you in understanding DNA?

The resource that has most helped me understand forensic DNA testing over the years has been my local lab analysts. There's simply no substitute for regular access to and asking questions of helpful and knowledgeable experts in the field of forensic DNA analysis. In addition, I'm fortunate to have had the opportunity to teach with and learn from some of the best forensic DNA experts in the nation at various meetings and conferences I've attended over the years. Aside from my interaction with experts, reading each edition of *Forensic DNA Typing* as well as other texts, journals, and web-based forensic DNA resources (such as STRBase) have greatly enhanced my understanding of DNA typing.

How much detail do you think that the jury needs regarding DNA testing?

I think Albert Einstein summed it up best when he said "Everything should be made as simple as possible, but not simpler." The jury needs just enough information to accomplish its assigned task—to determine whether or not the DNA test results in question are relevant, reliable, and persuasive in the context of a given case. In most cases this can be accomplished by the following: 1) demonstrate that the analyst is qualified and competent; 2) show that he or she works in a lab that follows established quality standards; 3) present a brief, basic explanation of DNA and the common biological materials in which it is found; 4) define essential terms to be used by the expert during his or her testimony; 5) provide an explanation of the evidence examined, including the results of any preliminary serological or trace testing; 6) offer a basic and brief explanation of the testing process, including the use of relevant controls; 7) explain that one or more genetic profiles were detected from the evidence and interpreted; 8) compare the profile(s) to the known standards and announce the testing outcome (i.e. match, partial match, inconclusive, exclusion); 9) provide the statistical frequency of the profile(s) developed; and 10) give a source attribution statement, if applicable and utilized by the testing laboratory.

What advice do you have to expert witnesses testifying on DNA evidence?

The best advice I have for any speaker, expert DNA witnesses included, is to know your audience and speak to them in their native tongue. A brilliant analyst can make a lousy expert witness when he or she fails to connect with the jury. Many times, this failure to connect stems from the fact that the expert testifies as if he or she is delivering a scientific dissertation to a room full of colleagues rather than teaching a classroom full of novices. Given the time and procedural constraints of the courtroom, the use of scientific jargon, industry acronyms, and excessively complicated explanations creates cognitive barriers for jurors that are difficult to overcome.

Expert witnesses should strive to testify in a user-friendly and ampliative *[enlarging a concept by adding to what is already known]* manner, only employing essential scientific terms after they have been properly defined. Additionally, more advanced concepts should be addressed only after their basic logical and factual predicates have been established.

How should expert witnesses prepare for their role in a trial?

Reviewing a case record before trial and preparing for trial testimony are not synonymous activities. A thorough review of the record is a necessary first step. However, being truly prepared for court requires additional preparation. This includes self-reflection on how to best equate, translate, and communicate scientific terms or concepts to the jury, mentally self-editing unnecessary terminology or redundant information, and anticipating the possible issues that may arise on both direct and cross-examination given the history of the evidence examined, the case data, and the test results.

Perhaps the most vital aspect of testimonial preparation is meeting with the prosecutor for a pretrial conference. If at all possible, this conference should be done in person, rather than over the phone. At this meeting, the prosecutor should ask the expert what issues or anomalies are present in the evidence, case data, or test results. The prosecutor should also review his or her anticipated questions and areas of inquiry with the expert and receive his or her feedback. It's also helpful for the expert to suggest issues or topics to highlight, deemphasize, or exclude. In addition, the prosecutor and the expert witness should collaborate to determine what, if any, visual aids may help enhance the jury's comprehension and retention of the evidence. Finally, an informal "dry-run" of the expert's direct examination or anticipated cross-examination during the pretrial conference may also help the prosecutor and the expert enhance their courtroom presentations.

How much time do you take with a DNA expert prior to court to prepare them for the courtroom?

To avoid wasting everyone's time, I make sure that I have done my homework before meeting with the expert witness. This means that I will have already reviewed the relevant documents and have isolated the issues, questions, and anomalies that are not explained by these materials. This allows me to focus on the unknown and the unclear rather than a time-wasting review of the test documentation that could have been done well ahead of the meeting. Ideally, by the time I meet in person with the expert, I will have a rough draft of his or her direct examination and a fairly good idea of the issues and questions that will be the subject of cross-examination. After the pretrial conference, I will typically revise and extend my outline based on the information learned during the meeting. If done efficiently,

preparation of the DNA expert can be done in about an hour in most cases. However, if the DNA results are *the* issue being litigated in the case, preparation may take multiple hours and a number of meetings.

Do you have a list of prepared questions that you use for your direct examination?

I have an outline of the facts I need to establish or the topics I need to cover, but I don't write out my questions. I believe that writing down questions verbatim locks you into a script that takes away the flexibility needed during witness examinations. Glancing down at a topic or an anticipated answer rather than a question gives me the flexibility needed to formulate a question that will elicit the desired information. If you get out of sync with an expert in court based on a prepared script, both prosecutor and expert are going to get lost and will look unprepared to the jury.

From your experience, is there anything unreasonable that the defense requests during discovery?

At times, some defense attorneys manipulate the discovery rules and make oppressive, burdensome, and unreasonable requests for the purpose of harassing the prosecution. One example of this is a request for all raw data generated incident to a lab's internal validation of a particular instrument or commercial DNA testing kit. Other examples include asking for all raw data generated from the profiles included in the lab's population database, and a request that all the electropherograms (which we provide on a CD) be printed rather than provided electronically—ostensibly because the defense expert does not own the necessary software. Another burdensome request is asking the government to produce the lab's procedures, protocols, audits, and analyst proficiency tests in each case, despite the fact that the same materials have been provided to the same attorney numerous times in the past on other cases. A popular request in the early 2000s was to ask for primer sequence information for the Profiler Plus and COfiler kits. The danger in not aggressively responding to these and similar requests is that judges who are not familiar with the subject matter may automatically order the state to produce the requested materials. In so doing, they wrongly assume the request is made in good faith and necessary to defend against the charges.

What are some of the common tactics that you have seen used by the defense attorneys and experts in challenging DNA evidence?

Defense challenges come in a variety of forms depending on the nature of the case and the significance of the DNA evidence. If the DNA results are clean and fairly unassailable, a popular defense tactic is to concede that the detected profile belongs to the defendant but that its presence at the scene was the innocuous result of a secondary or tertiary transfer. Along these same lines, when preliminary testing was not attempted or cannot determine the nature of the biological substance from which the profile was developed, the defense may concede that the profile belongs to the defendant, while asserting that it came from biological material whose presence at the scene or on the victim is easily and innocently explainable.

When the evidentiary context of the DNA leaves little room for innocuous explanations, the attacks turn to evidence handling, the testing process, and the interpretation of the

results. Depending on the jurisdiction, evidentiary chain of custody can be quite complex and involve many different people and locations, both inside and outside the lab. Attacking gaps in various links in the chain is an attack that is often available. Another popular defense tactic is to allege that the questioned DNA profile was the result of contamination from either inside or outside the lab (or both). This includes exploring prior instances of laboratory contamination on tests not related to the case being litigated. The failure of an analyst to strictly follow applicable standards, procedures, protocols, and internally validated methods is another topic that is a favorite of defense experts. (This is highly ironic because many such experts follow no written protocols at all in their own labs.) Also, a laboratory's past audit performances and an analyst's previous competency and proficiency test results make popular fodder for defense challenges.

An alternative interpretation of the analytical results produced by the state's lab is another routine defense challenge. Re-examining the raw data with computer software at a level beneath the lab's internally validated RFU threshold is a tactic routinely utilized by some DNA experts in the defense community. The purpose is to call into question the interpretation of the data by the government's analyst.

In addition, a defense expert's alternative interpretation of a DNA mixture is a fairly common occurrence. Such interpretations will frequently be used to argue that the defendant should be excluded from the profile(s) generated. Alternatively, it may be used to argue that alleles at certain loci should not be utilized in determining the population frequency of the genetic profile.

In cases involving cold hits, some defense experts argue that the only valid method of calculating the population frequency of the match is to use the database match probability rather than the random match probability.

Have these tactics been successful? Do you have any examples from specific cases that you would like to share where the defense successfully challenged DNA evidence?

By and large I would say that these tactics are ultimately not successful. I have yet to lose a case involving DNA evidence. Juries may have more information to sort through at the end of a case when a defense DNA expert is called, but a well-prepared state's expert will usually win the day in court. The danger for a defendant who calls a DNA expert is that such an expert will usually be obliged under cross-examination to concede a number of facts about the state's DNA evidence that are not helpful to his or her defense. Typically, I find it to be a net gain for my case if a defense DNA expert testifies at trial.

How has post-conviction DNA testing and the Innocence Project impacted the legal system and specifically your perspective on the criminal justice system?

To the extent that DNA evidence has been able to help exonerate the truly innocent, everyone has reason to celebrate. Although, at first blush, it may seem counterintuitive that finding and exposing past instances of injustice would enhance overall confidence in the criminal justice system, I think that will be the ultimate effect. Public perception of a fair and just process for correcting judicial mistakes will ultimately overshadow the tragedy of individual cases that resulted in wrongful convictions.

The cold case projects currently being operated by police departments, prosecutors, and crime labs across the country are an inverse but complimentary enterprise to that of the

Innocence Project. These projects take aim at delivering long-awaited justice from a different direction—providing closure for victims by cracking unsolved cold cases. These projects (including the one I oversee in Kansas City, which has resulted in over 100 cold case convictions with DNA to date) have disproven the old adage that justice delayed is justice denied.

My perspective on the system has changed to the extent that I have personally seen cold cases solved and justice achieved when police, prosecutors, and victims had abandoned all hope decades ago. To date, we have not had a DNA exoneration in Kansas City but have worked with the Innocence Project and other advocates on a number of cases that we found appropriate for post-conviction DNA testing.

What are some of the challenges faced by prosecutors with the new era of post-conviction DNA testing?

The most basic challenge faced by prosecutors is making an informed choice about when to agree with or object to a request for post-conviction DNA testing. This decision must take into account the nature and contextual significance of the evidence, its history since recovery, its suitability for particular testing methods, and the collateral facts and issues in the case. This can be a very difficult decision to make given the amount of information that must be located, reassembled, and carefully analyzed many years after a conviction.

Most of the disagreements I've had with the Innocence Project have stemmed from our differing opinions on the contextual significance of the evidence they want to test. An exclusionary DNA result is not synonymous with an exculpatory result. Due to the sensitivity of modern DNA technology, the number of items and samples typically collected at crime scenes, their unknown or equivocal history, and the nature of the sample, developing a genetic profile that does not match a convicted defendant in an old case may not be all that surprising. In short, depending on the context of the evidence, detecting an exclusionary profile may be much less probative than detecting an inculpatory one. Additionally, it only takes a single trial judge to determine that post-conviction DNA results are "exculpatory," whereas it takes a jury of 12 to unanimously agree that inculpatory DNA evidence should result in a guilty verdict. Thus, before stipulating to post-conviction DNA testing, prosecutors must fully realize the potential implications of this decision.

Other challenges in this new era include complying with new laws in some states that mandate the permanent post-conviction retention of evidence in certain cases; attempting to locate and determine the history and chain of custody of evidence that is the subject of a post-conviction testing request; finding old case files and transcripts; and searching for individuals from whom elimination samples are needed to compare against DNA profiles detected during post-conviction testing.

In many of my interactions with DNA analysts, I have found that they sometimes view the defense attorney and expert(s) as their enemies in a courtroom battle. What advice do you have to share in order to help DNA analysts appreciate the important role that a defense attorney has?

Science is a dynamic discipline. One of the hallmarks of good science is openness to criticism and revision. I believe this is also the hallmark of a good scientist. It's a rare fact that can

only support a single rational inference. It's the defense attorney's job to question, challenge, and attempt to refute inferences that support the state's case—including incriminating DNA evidence. Any analyst should take pride in the ability to withstand a well-prepared defense attack on his or her bench work and interpretive findings. It might even make them a better scientist by causing them to see old practices from new perspectives. To put it more bluntly, I'd remind analysts of the old adage, "What doesn't kill you makes you stronger."

DNA protocols, particularly with mixture interpretation, can change over time as the science improves and the laboratory embraces new techniques. How can these changes impact previous cases? Can you briefly explain *Brady* issues that may arise and why it is important for DNA analysts to understand the potential impact of protocol changes?

The *Brady* line of decisions requires the prosecution, as an aspect of due process and without a defense request, to disclose any information, either in its possession or in the possession of other governmental personnel, which is material to guilt or punishment. This includes information that is either exculpatory or has impeachment value. "Materiality" has been ruled to mean information that, had it been known at the time of trial, would have made a different result reasonably probable. The prosecution's good or bad faith in having knowledge of and disclosing such evidence is irrelevant.

Government laboratories, or those that contract with law enforcement agencies, should be aware that they are considered to be "state actors" for constitutional purposes when conducting tests used by the prosecution. This means that, irrespective of their good or bad faith, a lab's knowledge, actions, or inactions relative to *Brady* information in their possession is attributable to the prosecution. If a new interpretive protocol would cause a result-altering outcome on a DNA test in a pending case, the prosecutor must provide this information to the defense and the court. Prosecutors should make analysts with whom they work familiar with these obligations.

The United States Supreme Court, in *District Attorney's Office for Third Judicial Dist. vs. Osborne*, 129 S.Ct. 2308, 2320 (2009), recently made it clear that the prosecutor's constitutional duty of disclosure pursuant to *Brady* is a *trial right* enjoyed by the defendant. Thus, *Brady does not* control in the post-conviction context. Convicted defendants must now seek access to *Brady*-type information through the post-conviction procedures and legal precedents provided by the jurisdiction in which they were convicted. *Osborne* also held that the federal right to due process is only violated if a jurisdiction's post-conviction procedures offend a fundamental principle of justice or are fundamentally inadequate to vindicate a convicted defendant's substantive rights. Accordingly, prosecutors should make lab analysts with whom they work familiar with the post-conviction duties and obligations of disclosure imposed by the relevant jurisdiction.

Despite the *Osborne* decision, I would recommend that whenever interpretive protocol changes cast doubt on a convicted defendant's inclusion in a sample (mixture or not), or when a new protocol reduces the number of loci or alleles attributable to the defendant, thus increasing the frequency of his profile, the testing laboratory should immediately notify the prosecutor of the relevant facts. This will allow the prosecutor to make the appropriate decisions consistent with the legal and ethical dictates of the relevant jurisdiction and his or her responsibilities as an officer of the court.

INTERVIEW WITH THE ORANGE COUNTY (CALIFORNIA) DISTRICT ATTORNEY'S DNA UNIT

The Orange County (California) District Attorney's (OCDA) DNA Unit is comprised of experienced deputy district attorneys who possess expertise regarding the utilization of forensic DNA evidence to solve crime. They provide assistance and training to all of the county's law enforcement agencies. Members of the OCDA DNA Unit, which include attorneys Bruce Moore, Camille Hill, Scott Scoville, Jennifer Contini, Terry Cleaveland, and Tammy Spurgeon, work closely with local law enforcement investigators, prosecutors, crime lab forensic scientists, and OCDA local DNA database forensic scientists to effectively use forensic DNA technology and resources.

What is your role in the criminal justice process?

A prosecuting attorney has an obligation and duty to see that "justice is done." This obligation extends well beyond merely convicting the guilty. It includes ensuring that no one is unfairly convicted and that the innocent are exonerated:

> [The prosecutor] is the representative not of an ordinary party to a controversy, but of a sovereignty whose obligation to govern impartially is as compelling as its obligation to govern at all; and whose interest, therefore, in a criminal prosecution is not that it shall win a case, but that justice shall be done. As such, he is in a peculiar and very definite sense the servant of the law, the twofold aim of which is that guilt shall not escape or innocence suffer. He may prosecute with earnestness and vigor – indeed, he should do so. But, while he may strike hard blows, he is not at liberty to strike foul ones. It is as much his duty to refrain from improper methods calculated to produce a wrongful conviction as it is to use every legitimate means to bring about a just one (*Berger v. United States* (1935) 295 U.S. 78, 88).

As a lawyer how much do you want or need to know about the details of DNA testing?

As prosecutors, it is necessary for us to have at least a basic understanding of forensic DNA analysis so that we can spot potential issues and defenses. As the proponents who are seeking to introduce DNA evidence into trial, our level of understanding should include the ability to determine the probative quality of the DNA evidence, to explain the DNA evidence to a jury, and to determine the soundness and reliability of the DNA testing that was performed in any given case. We feel that a good understanding of forensic DNA evidence can empower a courtroom advocate. The jury intuitively grasps that you understand the science and are attempting to present reliable DNA evidence in a straightforward manner.

Of course, there is obviously a limit to the amount of forensic DNA knowledge an attorney can really possess, particularly those who present DNA evidence in a criminal case only occasionally. We have found it extremely helpful to have a core group of attorneys specially trained regarding the science and presentation of DNA evidence. These "DNA Unit Attorneys" possess more than basic knowledge regarding forensic DNA analysis, are well informed regarding DNA-related case law, and can assist other attorneys, forensic scientists, and local law enforcement investigators with complex legal forensic issues. Our DNA Unit Attorneys often act as liaisons between forensic scientists and non-DNA Unit prosecutors. They also help forensic scientists triage or screen cases prior to DNA analysis.

What resources have been most effective for you to learn the things you needed to learn regarding DNA?

In addition to reading scientific literature such as peer-reviewed journal articles and books, attending DNA conferences and seminars has been particularly helpful. Forensic scientists and academicians from all over the world have been generous with their time and knowledge. Joining scientific professional organizations has also been beneficial. Lawyers who present forensic DNA evidence in a courtroom setting should definitely be part of this human identification network.

How important is communication between forensic scientists and lawyers?

Forensic scientists, police investigators, and lawyers need to learn to effectively communicate with one another. One of the biggest issues is the lack of teamwork that exists between scientists, prosecutors, and investigators. A common refrain frequently heard from forensic scientists to lawyers is "You are not scientists!" As lawyers, we do understand and agree that our ability to understand complex DNA matters is limited. But, by the same token, scientists need to understand that they are not lawyers and cannot determine what is and is not relevant in a particular case.

Our worlds are colliding. We need expert witnesses. Many forensic scientists do not understand that a prosecutor's dual obligation is to be a zealous advocate and, at the same time, ensure fairness at every stage of the criminal proceedings. It is our duty, as prosecuting attorneys, to present the results of valid, correctly performed scientific analysis. Each time we intend to present DNA evidence to a judge or jury, we need to determine, among other things, if a problem occurred during the analysis of a sample; if the methodology is flawed; or if the analyst violated lab protocols. We cannot simply accept a DNA forensic conclusion expressed in a short paragraph of a one-page report at face value. Our exploration of these important issues as we prepare for trial should be welcomed by forensic scientists. A forensic scientist should be able and willing to explain and defend their scientific results leaving aside personal bias, ego, rancor, and defensiveness.

How much detail do you think that the jury needs regarding DNA testing? Are there things you can do during the trial to make sure the jury understands?

Generally it is wise to keep it simple, particularly if the evidence is not really being challenged. However, if it is anticipated that the challenges by the defense will be extensive, it will sometimes be necessary to educate the jury with more detail. We are finding, more and more because of the whole CSI effect, that jurors expect DNA evidence to be presented or they expect an explanation as to why there is not DNA. The jury does not need to know and understand as much DNA technology as a forensic scientist or attorney. The expert witness, however, should be prepared to respond to any issues the defense might raise in an articulate and straightforward manner. Our goal is for jurors to understand and believe that the DNA results presented at trial are reliable and reproducible.

With reference to DNA evidence, we find it helpful to train our attorneys to walk into court with a "can do" attitude. By this we mean to present your case as simply as possible, firm in the belief that the jury can and will understand what you are about to present. Even if you have the battle of the experts occurring, both attorney and forensic scientist should,

at all times, maintain a poised and confident demeanor. What prosecutors and forensic scientists often do not realize is that juries tend to stop listening after the first 20 to 30 minutes of technical DNA testimony. Jurors primarily make their decisions regarding DNA evidence both on the science presented and on the behaviors and attitudes of the expert witnesses and attorneys.

There is a group in Australia examining juror comprehension. They advocate discussing the basics of DNA and the non-contested parts of a case upfront before the trial actually starts. The jurors would then be in a better situation to handle the complexities that may arise later in the trial. Do you think something like that would work in the U.S. court system?

This is the equivalent of having the defense and the prosecution agreeing to a stipulation regarding forensic testing that would be read to the jury at an appropriate time. We are finding more and more that the defense attorney is willing to stipulate to DNA evidence, particularly in the cases where identity is not an issue or the evidence does not have significant bearing to the defense of a case. When DNA evidence, however, goes to the heart of a matter, the defense will rarely agree to a stipulation.

What advice do you have to expert witnesses testifying on DNA evidence?

The expert should keep it simple even when there are extensive challenges by the defense. Explanations need to contain enough detail to establish the credibility of the expert and the reliability of the science. The expert should thoroughly review the lab case file well in advance of a prospective trial date and should inform the prosecutor of any problems or issues that occurred during the analysis of a case. If errors did occur, the forensic scientists should immediately provide to the prosecutor a supplemental report and any corrective action documentation that was generated.

Forensic scientists should always strongly insist on a conference with the attorney who has requested that they testify prior to taking the witness stand. In a perfect world, the lawyers would be contacting the forensic scientists to arrange such a conference. But if the lawyer fails to do so, the forensic scientist should take the initiative and insist that a meeting take place. A forensic scientist should make clear during that meeting the facts or inferences that he or she is willing to state based on the scientific evidence. It is also important for the forensic scientist to gauge how much the prosecutor knows about DNA evidence.

The primary job of the forensic scientist at this stage is to defend the science. A forensic scientist should never confuse his or her duty to defend the science with the belief that they need to defend themselves. Too often, forensic scientists view an attack on the science as a personal attack. Falling into this trap will result in errors in judgment, incorrect statements, and will ultimately undermine a forensic scientist's ability to effectively respond to attacks on the science.

Remember, a forensic scientist's personal biases have no place in the analysis and interpretation of forensic DNA evidence. Simply put, a forensic scientist is not entitled to the luxury of expressing his or her feelings while performing their duties as an impartial scientist.

Do you provide feedback to expert witnesses after they have provided testimony?

Yes, in Orange County the prosecutors fill out an evaluation and send it to the lab where it is reviewed by their laboratory director and the scientist involved in the trial. In lieu of the form, we often talk to the forensic scientists personally regarding their testimony. Most often, the advice and feedback that we provide is well received.

From your experience, is there anything unreasonable that the defense requests during discovery?

Sometimes defense attorneys use a boilerplate discovery request that asks for everything "including the kitchen sink." Our standard discovery packet includes the report, all bench notes, a disk that contains electronic data, corrective action reports, and proficiency test results of the analysts who performed the testing.

Generally speaking, the defense is entitled to receive all discovery materials that would help them prepare their case. So, if the items requested are relevant, they should be provided. The law in California includes provisions that state that the defense must first approach the prosecutor for discovery materials. Upon occasion, discovery requests are improperly sent directly to the crime lab by the defense and are responded to by crime lab personnel without prosecutorial input. Without a thorough understanding of the law and a complete review of entire prosecution and investigation case files, a forensic scientist is not in the best position to make decisions regarding the release of discovery materials.

The discovery decision is first and foremost a legal and ethical decision for prosecutors. It might be easy for a forensic scientist to look at a discovery request and say "I understand what the defense means when they ask for 'all proficiency test reports.'" But a simple request for "all proficiency test reports" can become quite complex: Should the response include all proficiency test reports since the beginning of the lab's DNA testing? Should the response include all proficiency test reports for the last 10 years? Should there be any limitation in time? Should just the reports for DNA analysts be included? Should the response be limited to just the reports for DNA analysts who worked on a particular case? Should the reports include all written documentation from the entity that provided the proficiency test kits? Should the reports include a comparison of how local DNA analysts compared to analysts throughout the nation? Should all reports related to failed proficiency tests along with any corrective action reports generated also be included?

Depending upon jurisdictional rules and as a safe practice tip, forensic scientists should avoid placing themselves in the position of independently determining the legal and ethical requirements of providing discovery to the defense. It is the prosecution's duty and obligation to maintain a complete and thorough record of all discovery materials provided to the defense. Likewise, it is critical that forensic scientists maintain a complete log of all discovery materials provided to both the defense and prosecution. There is a very famous murder case here in Orange County that was reversed because of a discovery error. In that case, the crime lab indicated that they had provided a single-page report to the defense that was not provided to the prosecution. The report contained exculpatory evidence. On appeal, the defense claimed that they did not receive the report. Since the report was not provided to the defense through the prosecutor's office, no documentation existed that proved that the defense did indeed receive the report.

Communication between the prosecutor and forensic scientist is essential during the discovery process. The prosecution can be charged with the failure of turning over discovery materials to the defense even though they were not aware that the materials existed in the control of a third party such as a police agency or a crime lab. Many cold cases, some decades old, are being solved with DNA technology. The prosecution may be unaware that additional records exist besides a recent DNA report. Forensic scientists must take care to provide all notes and reports related to such cases even those stored in musty warehouses or basements that were written before the advent of computers and modern laboratory information management systems. Some discovery requests can indeed be quite burdensome to the crime lab by requiring the production or copying of voluminous materials. Forensic scientists should communicate these concerns to the prosecution. Prosecutors can help place limitations on the items requested or can arrange for a monitored viewing of the items by a defense expert at the crime lab.

Is it a problem to share electronic data files with the defense?

We see no problem providing electronic data files to the defense since any changes to the inherent data would be detectable by the crime lab forensic scientist. We have seen instances of some defense experts manipulating the appearance of the data by changing various parameters. However, with the aid of the forensic scientist, a prosecutor will be able to immediately recognize what was done and cross-examine the witness regarding the motives for the changes.

What are some of the common tactics that you have seen used by the defense attorneys and experts in challenging DNA evidence?

Challenges to DNA evidence fall into the following categories: contamination of the evidence at the crime scene or lab; erroneous determination of allele peak calls; improper interpretation of data especially with low levels of DNA or mixtures; improper frequency calculations; improper methodology used to conduct validation studies, set stutter limits, set analytical thresholds, and set stochastic thresholds; forensic scientist bias both pro-prosecution and pro-defense; analyst errors such as failed proficiency tests or failure to follow lab protocols; failure to follow basic scientific method; and lack of transparency.

One of the latest tactics we have seen: the defense will call multiple experts to the stand but limit their area of testimony. For instance, the defense attorney will call an "on the fringe expert" to the stand who will make questionable calls by including spikes, stutter, and obvious noise as allele peak calls. Then they will call a human geneticist well versed in both the interpretation of DNA data and statistics to only testify regarding frequency calculations. This tactic is easy to defuse by having the geneticist recalculate frequencies using only valid allele calls while on the stand.

"Dueling Guidelines" is another tactic. Sections of the manufacturer's guidelines and validation studies are put up on a screen in the courtroom next to copies of the laboratory's interpretation guidelines to highlight the differences. Also, defense experts will sometimes change the lab's threshold values or narrowly evaluate data ignoring relevant sections of the lab's protocols.

The forensic scientists need to be strong enough to say that they did something a certain way because of their laboratory's validation studies. With these types of challenges, it is important that a forensic scientist feel comfortable and knowledgeable with the relevant

laboratory protocols and validation studies. A forensic scientist exhibiting a lack of confidence or understanding of their own protocols and validation studies will undermine his or her credibility with the jury.

One of the challenges with forensic DNA analysis today is mixtures…

One major area where forensic scientists could benefit from additional study and training is the statistical interpretation of complex mixtures. Mixture interpretations are often difficult and many forensic DNA analysts do not yet have the skill set to properly interpret complex mixtures. In California, the presentation of DNA results during a courtroom proceeding must be accompanied by a statistical frequency. It is paramount that forensic laboratories provide additional statistical training for their DNA analysts. It would also be beneficial to have at least one person in a lab who is an expert in the field of statistics.

How are new mixture interpretation guidelines going to impact you in cases you are dealing with?

Forensic scientists may not realize that a revision of their lab's mixture interpretation protocols may also necessitate a review of all previous mixture interpretation results and conclusions. We agree with other prosecution experts that whenever interpretive protocol changes cast doubt on an individual's inclusion in a sample or when new DNA interpretation guidelines reduce the number of loci or alleles attributable to that person, thus increasing the frequency of the profile, the testing laboratory should immediately notify the prosecutor of the relevant facts. Timely notifications will allow the prosecutor to make the appropriate decisions consistent with the legal and ethical dictates of the relevant jurisdiction and his or her responsibilities as an officer of the court.

So how do you handle this balance then for the future? The challenge is that science is always going to improve and if the law doesn't want it to improve then that impedes the science.

Improvements in science are always welcome. There needs to be an understanding, however, that forensic scientists have a duty and obligation to notify prosecutors whenever scientific improvements occur that would change a previous scientific conclusion.

I just think that the forensic community is in a very precarious situation with how mixtures are being handled right now…

One suggestion for a forensic lab would be not to allow new DNA analysts to interpret complex mixtures. Perhaps a lab should have a core team of 3 or 4 highly experienced DNA analysts that work together to interpret each complex mixture and thoroughly document all of their assumptions and conclusions.

You will notice that the new SWGDAM guidelines are very clear that you need to document all of your assumptions. There is a reason that requirement was put in there.

It does not help if all findings and conclusions are not thoroughly detailed in a lab report or notes. When a forensic scientist interprets a complex mixture, care needs to be taken to document every aspect of that interpretation. Another scientist examining the notes should be able to see and understand the basis of all of the mixture interpretation decisions.

That comes down to a training issue then. Why is there that kind of disparity in knowledge between the new analyst and the experienced analyst? Why can't there be better training so the newer analyst can come up to speed faster on mixture interpretation?

Better training would indeed help, particularly the inclusion of more statistical training. Acquiring knowledge and experience takes time. Unfortunately, there is no substitute for the experience a forensic scientist will gain from analyzing thousands of cases and samples. Since we know that complex mixture interpretations are based, in large part, on the experience and training of the analysts performing the interpretations, the lab needs to ensure that all persons performing complex mixture interpretations have achieved an appropriate level of training and competence in this area.

How has post-conviction DNA testing and the Innocence Project impacted the legal system and specifically your perspective on the criminal justice system?

California law allows for post-conviction DNA testing pursuant to Penal Code section 1405. Moreover, Orange County District Attorney's Office has an Innocence Review Panel that reviews all cases where a defendant believes his/her case deserves another review when forensic evidence exists. With both of these vehicles in place, the criminal justice system is better equipped to ensure that justice is served.

On those types of cases where there is testing on someone who says they are innocent, do you keep any stats on how many times that DNA comes back matching?

We have tested three or four cases as part of our Innocence Review Panel and each time the DNA results inculpate the defendant. As prosecutors, we are advocating for a certain position. But we also have the dual responsibility to ensure that justice is served.

We had a recent case that involved the kidnap and sexual assault of two young girls from the 1980s. The case against the defendant was extremely compelling even without DNA evidence. Nevertheless, we agreed, as part of the Innocence Review Panel, to perform DNA testing. The DNA profile from the crime scene evidence matched the defendant.

INTERVIEW WITH BRADLEY BANNON, J.D., ON A DEFENSE ATTORNEY'S PERSPECTIVE

Mr. Bannon is a criminal trial defense attorney for the law firm of Cheshire, Parker, Schneider, Bryan, and Vitale in Raleigh, North Carolina. He has been practicing law for 13 years and has helped draft legislation to strengthen the right of the accused to pretrial discovery in criminal cases in North Carolina. He was instrumental in using DNA evidence to help exonerate the Duke lacrosse players of false accusations of rape in 2006 and 2007.

What is your role in the criminal justice process?

To provide independent and zealous representation to people accused of crimes, and to give as much meaning as possible to the presumption of innocence and the burden of proof that the government must meet before it may limit or deprive a person of freedom or the full rights and privileges of citizenship.

There are several phases of that representation. First, you review the prosecution's evidence and independently pursue as much evidence as you can and should under the particular circumstances of each case. After evaluating all of the evidence in light of your practical and legal knowledge and experience, you provide your best advice to the accused about what he or she should do, whether that involves going to trial or exploring resolutions outside of trial, such as a plea agreement. Finally, if the case ends up in a contested forum, whether a trial or sentencing hearing, you do your absolute best to make sure the prosecution is put to its constitutional burden in that forum.

You were defense counsel in the Duke rape case. What lessons were learned from this experience that you would like to share?

The biggest lesson was learning that you should never accept at face value what you see in a report that you receive from an expert. That does not mean the expert is necessarily lying or trying to pull the wool over anyone's eyes, though that certainly does happen. The reason for not accepting a forensic scientific report at face value is that, whatever science is being applied, it is being applied by human beings. The scientific standards and laboratory protocols are being followed—or *not*—by people, and *people* are the ones reporting the conclusions, not machines.

At a time when the need to confront forensic evidence issues effectively in the courtroom is directly proportionate to the explosion of the portrayal of such evidence in American media and pop culture, it is more important than ever for lawyers to know the fundamentals of the forensic sciences. You can't just put your head in the sand and approach cases as you would have in the 1980s or even in the 90s. Advocates for the accused must evolve with the evolving types of theories, themes, and evidence presented in prosecutions of serious crimes.

Partly because of advances in the forensic sciences, and partly because jurors increasingly expect to see "CSI" evidence in criminal trials, forensic evidence such as DNA analysis is increasingly prevalent in criminal courtrooms. To the extent it is being presented through "experts" associated with law enforcement, such as those who work in state and local crime labs, it carries a tremendous presumption of credibility with jurors. But because those experts are human beings who are just as prone to human frailty as the rest of us, defense lawyers have an increasing duty to understand the forensic sciences being used in the courtroom to try to convict their clients, and, when those sciences are not being used appropriately or accurately, to make that known to the court and jury.

For the most part, over my first decade of defense work, I never had occasion to question a DNA report. I treated DNA evidence like I treated any other type of evidence: if it was consistent with my understanding of the facts and theory of the defense, I saw no need to question or challenge it. But if the accused is adamantly telling you "I did not do this, I wasn"t there, it cannot possibly be me…," or if the DNA evidence is otherwise inconsistent with the theory of defense, then it is a defense lawyer's duty and responsibility to become familiar with the fundamental concepts of DNA analysis in order to review that evidence.

That first happened with me when the Duke case came along. I absolutely believed the accusations were false, and I had zero confidence in the prosecution's portrayal of the evidence. As a result of having the lives of innocent people in my hands, and anticipating that the prosecutor would try to use DNA evidence to convict them, I felt that I needed to

learn something about DNA so that I could determine what that evidence in the case truly showed, whether the prosecution's portrayal of it was complete and accurate, and, if not, how we could make the truth be known.

There is a school of thought among some lawyers that you can just ship off the DNA data in a case to an expert and let her tell you what it means. The problem with that approach is that the expert doesn't usually know *all* of the facts of the case, what the applicable laws and rules of evidence are, and how *all* of the facts, law, and science interact in the particular case. Just as importantly, the expert is not responsible for presenting her own testimony in the courtroom or for cross-examining the opponent's experts. Only the lawyers can do that. So if there is a flaw in your opponent's portrayal or presentation of DNA evidence *in the courtroom*, but all you have done with DNA up to that point in the case is to rely on an expert to tell you about it, and you have not also learned the fundamentals of DNA analysis yourself—if, in effect, you have allowed yourself to be given a fish rather than taught how to fish—then when the time comes *in the courtroom* to effectively confront questionable DNA evidence, you won't know how to do it. For example, if you're dealing with a lab analyst who violated the lab's own protocols, or applied them selectively, to reach certain conclusions in a case, you won't be able to effectively engage the expert on those very serious credibility issues in the courtroom, because you won't know what protocols *are*, you won't know why they exist, and you won't know how they are applied.

But don't get me wrong: familiarizing oneself with basic concepts of the forensic sciences does not and should not replace meaningful consultation with independent experts in those sciences. The point here is that such consultation is no substitute for the lawyer knowing all of these concepts, because he or she is the only one who is going to know all of the law, all of the facts, and how they interact with the science in the case, and the lawyer is the only one who can do anything about it in a courtroom on behalf of the accused.

What resources have been helpful to you in understanding DNA?

I began learning fundamentals about DNA mostly through your book, *Forensic DNA Typing, Second Edition*, and a couple of very informative articles in *The Champion*, the magazine published by the National Association of Criminal Defense Lawyers. Though I did not have any kind of scientific background, the book and articles were accessible to me, which means they should be accessible to everyone.

As I was reading the materials, the first light that went off in my head was that, while DNA analysis is widely accepted as the gold standard of forensic science, its application is much more open to human error than I had ever known. When I started out, I thought you simply shoved samples into a machine, and it spit out a report—and that was it. In reality, DNA testing—especially regarding samples that are weak or degraded, or contain DNA mixtures, which many evidentiary samples do—involves a lot of subjective interpretation of the electropherograms and what an analyst identifies as a "true peak." That's one of the main reasons I learned why it is so important for analysts to have and consistently follow protocols.

How much do you want or need to know about the details of DNA testing?

I think the most important thing to know is that there are between 13 and 16 areas (loci) that are examined for the presence of genetic markers, or "alleles" or "peaks," and

those peaks are portrayed in charts known as "electropherograms." It is important to be able to read electropherograms and to know the protocols of the lab that analyzed the samples, interpreted the charts, and produced the final report. It's also important to know concepts surrounding the difference between what is or might be a true DNA marker and what is or might be an "artifact," such as "pull-up." It's important to know that the strength of the marker, or the height of the peak, can mean a number of different things—that a sample is strong, or that more than one person's shared DNA characteristics may be in a mixture, etc.

Recognizing when a mixture is present in an evidentiary sample, or when the sample is weak or degraded, is also very important and raises all sorts of questions about how the DNA analyst chose to interpret and report (or not report) the sample. Again, there is no big DNA machine that intakes samples and outputs reports, so it's important to portray the entire process for what it is: one that involves human beings at all levels, from sample handling, to sample processing, to sample interpretation, to final reporting. Since reporting is driven by interpretation, and interpretation is necessarily subjective, that's why it's so important to know the lab's protocols.

Simply put, the protocols are the rules, and a core concept of scientific analysis is whether you apply the same rules and standards consistently in your work. If you choose to deviate from those rules on a case-by-case basis, and if you have no credible scientific explanation for doing so, that's not really science at all. Knowing whether an analyst followed established protocols to reach and report a conclusion is as important as knowing the conclusion itself. If there is not a credible scientific explanation for a deviation from protocols, it obviously calls into question the credibility of the conclusions. In the Duke case, we were able to effectively attack the credibility of a DNA lab's work based, among other things, on its violation of multiple protocols in reaching and reporting its conclusions.

Another lesson I learned in the Duke case is that language is very important, and the words an expert chooses to use in a final DNA report do not always accurately or completely reflect the underlying data. To determine the accuracy and appropriateness of the reporting language, the lawyer needs to learn some of these fundamental concepts about DNA analysis and reporting. The lawyer should also explore the lab's protocols (if any) and the expert's own customary practice regarding what analyses make it into the final report and what language she uses to characterize the analyses and conclusions. Is she always consistent? Does everyone in her lab apply the same rules and use the same language? Or do they sometimes report (or decline to report) certain things, or use different phrases to explain the same thing, or the same phrase to explain different things? If so, why would that be?

For example, I've seen reports from different DNA analysts in the same lab that use "match" versus "cannot be excluded" or "partial match" versus "cannot be excluded." The term "partial match" seems very compelling. What does it mean to say "cannot be excluded?" How many other people may also not be excluded? Does one profile truly "match," or even "partially match," another? If there is a "match" between a suspect's DNA sample and an evidentiary DNA sample at three or four of the 13 or 16 loci, should that be reported as any kind of a "match" at all? And how many other people could be a "match" to that sample in the lab's profile database? Or in the world? Should you even be able to introduce DNA evidence against a defendant if there was only a "match" at a single locus?

When talking to other defense attorneys, do you find that most of them just give up when they hear that a case involves DNA?

It can be daunting to face any new complex subject matter, particularly when it involves scientific analysis. DNA is scary to a lot of people. It was scary to me. When I had to learn about DNA for the Duke case, it's not like I was excited about it. I didn't think to myself, "Wow! This is a great opportunity for me to learn something new!" I was thinking, "Wow, I got an English degree in college and took no science classes." This is a subject matter that people go to school, and graduate school, and post-graduate school, for years and years to learn. If you're like me and have no real talent or educational background when it comes to math or science, then learning about DNA is a challenge. That said, I'm living proof that, with the right source materials and the commitment of time, you can learn the basics necessary to provide effective assistance of counsel to accused people in cases involving DNA.

What do you normally request for discovery when DNA evidence will be presented? Have you had any problems obtaining the case folder materials, the electronic data, or the Standard Operating Procedures for the DNA analysis?

In the Duke case, I asked for everything I could get, from underlying data to protocols to the DNA profiles of the lab technicians and analysts. One of the labs resisted a little bit, saying it would be costly, but the presiding judge ultimately issued an order requiring the lab to comply. In addition to receiving the underlying data with the exculpatory evidence that had not been mentioned in the final report, we got the lab's protocols, which allowed us to discover and highlight that the lab had violated them.

Here in North Carolina, in light of several very high-profile post-conviction exonerations of people who had been convicted and sent to prison (and even death row) because the defense had not been provided with information that the U.S. Constitution requires to be disclosed to defendants before trial, our legislature fundamentally changed the state discovery laws in 2004 to require state prosecutors to provide criminal defendants before trial with everything in the prosecution and law enforcement files, including the results and underlying data regarding any testing done in the case.

That said, defense lawyers here and all over America have historically faced resistance to discovery requests from labs, particularly when the requests seem aimed at probing the credibility of the lab's work and conclusions, and particularly when the lab is a creature of law enforcement. When the lab is essentially an arm of law enforcement, as are many crime labs throughout America, including ours in North Carolina, there is a real danger for development of a lab culture that subordinates scientific mentality to a team mentality, with the team being law enforcement and prosecution, not the scientific method. As long as that occurs in theory and practice, at best, the credibility of the lab's work and conclusions will always be in question. At worst, innocent people will be accused and convicted of crimes they did not commit, the real perpetrators of those crimes will not face justice, and the victims of crime will be further victimized.

Do you have sufficient access to experts to review the material when DNA evidence will be presented?

Yes, we have a solid and talented criminal defense bar in North Carolina and lawyers who are always willing to share their knowledge and experience with DNA in the

courtroom. We also have a database of recommended experts and an active listserv where lawyers regularly post and respond to requests for recommendations for experts to help on a case. Some people may want a DNA expert who can sit down with them for a day or two and explain the basics of DNA analysis. Others may want someone to run independent sample analyses and testify about them in court. Others may want someone who can help interpret the data. I've been very satisfied with the experts who have agreed to work with me over the years.

What are some common tactics that you have used in challenging DNA evidence?

Contamination—and I'm including secondary and tertiary transfer arguments under this umbrella—is, of course, the most commonly known method of attacking DNA evidence. Contamination can happen at the scene of collection, or in the laboratory, or somewhere in between. Preservation of evidence and the chain of custody are important. It's important to make sure the evidence wasn't comingled with other evidence that might have impacted the credibility of the analysis and results.

But contamination and chain-of-custody arguments are only the tip of the iceberg of potential challenges to DNA evidence. Beneath the surface are all of the types of challenges that really require the lawyer to know the fundamentals of DNA analysis itself. Did the lab follow acceptable standards of DNA analysis? Did it follow its own protocols? Is the lab applying those standards and protocols consistently or selectively? For example, why do you call a peak below 150 RFU as a true allele for one purpose, or in one case, but not for another? If there are such internal inconsistencies, do they usually inure to the benefit of one side's theory of the case? If so, is that evidence of bias?

Other factors that impact the credibility of a DNA expert's work are the amount and quality of his training and experience. Lawyers should seek the expert's CV and review the training and experience sections. Obviously, not all experts are the same.

The goal is being able to identify, where it exists, the weakness or flaw in the expert's analysis, reporting, and conclusion. If there were steps along the way when the DNA analyst had to make a judgment call, and if that judgment call went consistently in favor of the prosecution—especially if those judgment calls were inconsistent with other work that expert had done, either within the same case or in other cases—that is important for the judge and jury to know when you're asking them to reject the analyst's conclusion or the prosecution's portrayal of it.

This all returns to the main point: when the evidence supports challenging a DNA expert's conclusion, you must demystify the process and demonstrate to the jury that DNA analysis is a subjective human exercise. It is not foolproof. It is subject to human nature, human error, and human bias. So is any conclusion reached and reported by a DNA expert, and so is the expert's courtroom testimony.

What advice do you have to expert witnesses testifying on DNA evidence?

Always remember that your sole allegiance on the witness stand is to the scientific method and to the facts. You are not an advocate. If you are a defense expert, you are not carrying the torch for the defense's theory. If you are the state's expert, you are not carrying the torch for the theory of prosecution. In the courtroom, you are first, foremost, and only a scientist, and your duty is to explain your analysis and conclusions truthfully, factually, and

consistently with the scientific method, in language that is as basic and accessible to normal people like me as possible.

Which brings me to my second piece of advice: understand that you are not lecturing to your professional colleagues. You are speaking to people who will likely have little or no experience in your particular area of expertise, but you must effectively share that expertise with them without their eyes glazing over. Speak in language that is accessible but not condescending to them.

How has post-conviction DNA testing and the Innocence Project impacted the legal system and specifically your perspective on the criminal justice system?

I'm not sure anything has had a greater impact on the criminal justice system since I've been practicing. It seems like people are being released from prison on a weekly basis these days based on post-conviction DNA exoneration. Without question, the American public now knows that police and prosecutors, even in good faith, sometimes pick—and juries sometimes convict—the wrong people. Though skepticism of criminal accusation has long been embodied in the presumption of innocence and the requirement of proof beyond a reasonable doubt for conviction, collecting all of the most eloquent speeches in the world about *why that skepticism is right* does not add up to the power of one, let alone dozens, let alone hundreds, of stories about people who served 5, 10, 15, sometimes up to 30 years for crimes they did not commit. Because many of the original convictions in the DNA exoneration cases were based on mistaken eyewitness identification, it has also given stakeholders in the criminal justice system—judges, prosecutors, defense attorneys, law enforcement—a healthy concern about such evidence and has led to reforms in many eyewitness identification procedures across the country. As importantly, in addition to exonerating the innocent, post-conviction DNA analysis has also resulted in the identification, apprehension, and conviction of the actual perpetrators of the crimes. After all, forensic science is not just about making sure that the wrong guy does not get convicted: it's about making sure the right one does.

The sad thing to think about is this: how many other wrongful convictions—whether based on eyewitness testimony or other circumstantial evidence—did *not* involve the collection of physical evidence that can now be DNA-tested? Without question, many people are sitting in prison today based solely on such evidence, proclaiming their innocence as loudly as those who have been exonerated by DNA evidence, but without the ability to test their convictions in a similarly decisive fashion. I don't think anyone could intellectually honestly say that the only people who have been wrongfully convicted of serious crimes in America are just those people who have been exonerated through post-conviction DNA analysis.

How do you prepare for cross-examination of a DNA expert?

I suppose if I am in the position of cross-examining a DNA expert, I've already gone through the process of determining that there is a factual and legal need and basis for challenging the expert's conclusions and/or the prosecution's characterization of the DNA evidence. If that's the case, I brush up on the concepts of DNA analysis that I first learned while working on the Duke case, and I try to find out the extent to which they've changed. As important as it is to learn the fundamentals of DNA analysis in the first place, it is equally important to update that knowledge.

Then I basically apply the rest of the approach I've talked about in this interview. Depending on what kind of challenge I'm mounting, I focus on the part of the expert's process and conclusions that will support that challenge, and I design a cross-examination around those issues that is accessible to the jury. But no script is ever followed in a courtroom. That's why it's important for the lawyer to know the law, the facts, and the fundamentals of the science. Another reason is that some experts, consciously or subconsciously, become advocates for the prosecution, and you have to be able to recognize the difference between science and advocacy when an expert is testifying. If I suspect before trial that I am facing such an advocate-expert, or I have some other reason to doubt the expert's credibility or objectivity, I will seek input from other lawyers who have had cases involving that expert, and I will seek and review the expert's previous work and testimony.

While it's never my intent during cross-examination of a DNA expert to get into a "contest" of who knows more about DNA, my goal always is, as it is with any forensic science expert, to be able to have a meaningful dialogue with the expert about the case-specific issues in her area of expertise and to make the points I set out to make—or even a few new ones, should the opportunity present itself—without being intimidated by the knowledge gap.